

MECH 151: Technical Design — Prototyping

Unit Competency & Knowledge Assessment Questions

Drafted from the Unit Reference Slide Deck and syllabus schedule — 5 questions per unit, answer key included

Each unit below carries five questions blending recall (multiple choice / identification), explanation (short answer), and applied reasoning (scenario, calculation, or design judgment), matching the four unit quizzes on the syllabus (Weeks 7, 10, 13, 16). A single answer key with brief rationale for every question, grouped by unit, appears at the end of this document.

Unit 1 — CAD Fundamentals (Basic)

Weeks 4–7. Assessment: Unit 1 Quiz, Week 7.

Question 1 — Multiple Choice

Which sequence correctly describes the dimensional progression used to build a solid parametric model?

- A. 3D Solid → 2D Profile → 1D Line → 0D Point
- B. 0D Point → 1D Line/Curve → 2D Polygon/Profile → 3D Solid/Volume
- C. 1D Line → 0D Point → 3D Solid → 2D Profile
- D. 2D Profile → 3D Solid → 0D Point → 1D Line

Question 2 — Short Answer — Constraint States

A sketch still has some geometry that can move freely when dragged, even though several dimensions have been added. Is this sketch under-defined, fully defined, or over-defined? What general rule should be followed before adding dimensions, and why?

Question 3 — Applied — Orthographic Projection

In third-angle projection (the U.S. standard), where is the top view placed relative to the front view, and where is the right-side view placed? A student places the top view below the front view on a drawing — is this correct convention for a U.S. drawing, and what would you tell them?

Question 4 — Applied — Measurable Properties

Explain the difference between a 2D profile's centroid and a 3D solid's center of gravity. Then explain why assigning an incorrect material density in the CAD software produces a wrong mass moment of inertia even when the model's geometry is completely accurate.

Question 5 — Short Answer — Design Philosophy

Two students model the same bracket using completely different feature trees, yet both receive full credit. What criteria actually determine whether a parametric model is judged “correct” in this course, given that there is no single right feature tree for a given shape?

Unit 2 — CAD Advanced

Weeks 7–10. Assessment: Unit 2 Quiz, Week 10.

Question 1 — *Multiple Choice — Section Views*

A cutting plane for a section view passes directly through a rib on a part. According to standard drafting convention, how should the rib appear in the section view?

- A. Hatched, exactly like the solid material around it
- B. Not hatched — ribs, webs, fasteners, shafts, and pins are conventionally left unhatched even when the cutting plane passes through them
- C. Shown with a dashed hidden-line outline only
- D. Omitted from the section view entirely

Question 2 — *Applied — Auxiliary Views*

When is an auxiliary view needed instead of relying on the six principal (orthographic) views, and what does the reference-plane method accomplish when constructing one?

Question 3 — *Short Answer — Thread Notes*

Decode the thread callout “1/4-20 UNC-2A.” Identify what each part of this designation specifies.

Question 4 — *Applied — Fits*

A shaft-and-hole pair is specified as $\varnothing 25$ H7/p6. Is this a clearance or interference fit? Contrast it with $\varnothing 25$ H11/c11, and describe what physically happens when you try to assemble each pair.

Question 5 — *Multiple Choice — Tolerancing Notation*

Which notation denotes a basic (theoretically exact, GD&T-referenced) dimension, as opposed to a reference dimension shown for information only?

- A. A value shown in parentheses, e.g. (25)
- B. A value boxed in a rectangle, e.g. [25]
- C. A value followed by $\pm$, e.g. 25 ± 0.1
- D. A value with the $\varnothing$ symbol, e.g. $\varnothing 25$

Unit 3 — Design for Assembly (DFA)

Weeks 11–13. Assessment: Unit 3 Quiz, Week 13.

Question 1 — *Short Answer* — *Mates & DOF*

A free rigid body has 6 degrees of freedom. Explain how a concentric mate plus a coincident mate, applied together to two cylindrical parts being assembled, reduce those degrees of freedom.

Question 2 — *Applied* — *Joining Method Selection*

You are designing a 3D-printed plastic enclosure that must be assembled and disassembled multiple times for maintenance, using threaded fasteners into printed bosses. What minimum thread engagement depth should you plan for relative to the screw diameter, and why might a heat-set threaded insert be preferred over threading directly into the printed plastic?

Question 3 — *Applied* — *Joining Method Selection*

A design needs a joint that flexes repeatedly to allow opening and closing, without a separate hinge pin or hardware. Which joining approach fits this need, what is this joint type called, and which two plastics are specifically noted as tolerating repeated flexing?

Question 4 — *Short Answer* — *DFA / BOM*

Using a housing / cover plate / gasket / fastener BOM as a reference, explain why both “reducing part count” and “preferring off-the-shelf components” are DFA principles that reduce assembly cost.

Question 5 — *Applied* — *Sourcing*

Why does importing a fastener model from an online repository like McMaster-Carr (with a real part number) strengthen a bill of materials, compared to modeling a generic bolt from scratch? What limitation do imported models typically have once brought into your assembly?

Unit 4 — Design for Manufacturing (DFM)

Weeks 14–16. Assessment: Unit 4 Quiz, Week 16.

Question 1 — Short Answer — GD&T Families

Sort the following GD&T characteristics into their correct family — Form, Orientation, Location, Profile, or Runout — and note which families require a referenced datum: flatness, perpendicularity, position, profile of a surface, circular runout.

Question 2 — Applied — Feature Control Frame

Decode the feature control frame “ $\varnothing$ 0.2  A B C.” What does each element specify, and in what order do the three datums establish the datum reference frame?

Question 3 — Applied — Modifiers

Explain the difference between the MMC and LMC modifiers, and why a hole drilled larger than its worst-case (MMC) size is allowed extra (“bonus”) positional tolerance.

Question 4 — Applied — Process & Defects

A printed part comes off the printer with its edges curling upward and lifting away from the bed. Name this defect, its cause, and at least one fix. Then explain why this same design concern (cooling-driven distortion) would not apply if the part were laser-cut instead.

Question 5 — Applied — Safety & Materials

Why must PVC (or other chlorinated plastics) never be cut on a laser cutter? Also name one PPE requirement for operating rotating shop equipment such as a drill press or lathe.

Answer Key

Unit 1 — CAD Fundamentals (Basic)

Q1.

A. 3D Solid → 2D Profile → 1D Line → 0D Point

B. 0D Point → 1D Line/Curve → 2D Polygon/Profile → 3D Solid/Volume

C. 1D Line → 0D Point → 3D Solid → 2D Profile

D. 2D Profile → 3D Solid → 0D Point → 1D Line

B — a sketch builds up from points, to lines/curves, to a closed 2D profile, which is then given a third dimension (via extrude, revolve, sweep, or loft) to become a 3D solid.

Q2.

Under-defined — geometry that can still move has not had its position or size fully pinned down by relations and dimensions. The rule is to add geometric relations (coincident, parallel, tangent, concentric, symmetric, etc.) before adding dimensions, because relations capture the design's intended behavior (e.g., “always centered,” “always tangent”) more robustly than dimension values alone, and reduce the number of dimensions needed to fully define the sketch.

Q3.

In third-angle projection, the top view goes above the front view, and the right-side view goes to the right of the front view. Placing the top view below the front view follows first-angle projection (the convention used in much of the rest of the world, not the U.S. standard), so the student's layout would be non-conforming for a U.S./ASME Y14.5 drawing — they should move the top view above the front view (or explicitly label the drawing as first-angle if that convention was intentional).

Q4.

A centroid is the geometric center of a 2D area (independent of material); center of gravity is the mass-weighted balance point of a 3D solid, which depends on both geometry and how mass (density × volume) is distributed through it. Because mass moment of inertia is computed from the mass distribution, an incorrect density scales or skews the calculated mass at every point in the solid, so the result is wrong even though the geometry itself is perfect — this is the “garbage in, garbage out” problem: correct geometry cannot compensate for incorrect material properties or a non-watertight solid.

Q5.

Models are judged on: (1) whether they capture the design intent (the reasoning behind the shape, via relations, dimension choices, and feature order/references), (2) whether they rebuild cleanly without errors, and (3) whether they remain easily editable. A model is not judged by matching one expected sequence of features — effectively infinite valid feature trees can produce the same shape.

Unit 2 — CAD Advanced

Q1.

A. Hatched, exactly like the solid material around it

B. Not hatched — ribs, webs, fasteners, shafts, and pins are conventionally left unhatched even when the cutting plane passes through them

C. Shown with a dashed hidden-line outline only

D. Omitted from the section view entirely

B — ribs, webs, and similar thin support features (along with fasteners, shafts, and pins) are conventionally never hatched in a section view even when the cutting plane passes through them, because hatching them would misleadingly suggest they are solid, uninterrupted material and would clutter/obscure the drawing.

Q2.

An auxiliary view is needed when a surface is inclined relative to all principal viewing planes, so none of the six standard views shows that surface in its true shape/size — a primary auxiliary view is projected perpendicular to the inclined surface to reveal its true shape. The reference-plane method transfers measurements (depths/widths) across the fold line from an adjacent view into the auxiliary view, letting you construct the auxiliary view accurately without re-measuring from the 3D part.

Q3.

1/4 = major (nominal) diameter, 1/4 inch. 20 = threads per inch (pitch). UNC = Unified National Coarse thread series. 2A = class of fit (class 2, external thread — the “A” denotes external; “B” would denote internal). Together the callout fully specifies an external, coarse-pitch, 1/4-inch-diameter thread with a class 2 fit.

Q4.

Ø 25 H7/p6 is a close/interference fit — the shaft is sized so there is effectively zero (or negative) clearance; assembly typically requires pressing or light tapping, and the parts will not spin or slide freely once mated. Ø 25 H11/c11 is a loose/clearance fit — a visible gap exists between shaft and hole, giving a free-running clearance fit that the parts can be assembled and rotated/slid by hand with no force needed.

Q5.

A. A value shown in parentheses, e.g. (25)

B. A value boxed in a rectangle, e.g. [25]

C. A value followed by ±, e.g. 25 ±0.1

D. A value with the Ø symbol, e.g. Ø 25

B — a boxed value like [25] is a basic dimension: a theoretically exact value used as the reference for a GD&T tolerance zone, carrying no tolerance of its own. A value in parentheses, like (25), is a reference dimension provided for information only and is not toleranced or used for inspection.

Unit 3 — Design for Assembly (DFA)

Q1.

A concentric mate aligns the two parts' cylindrical axes, removing the two translational degrees of freedom perpendicular to that axis (and constraining rotation about the two axes perpendicular to it), leaving the parts free to slide along and rotate about the shared axis. A coincident mate (e.g., mating a face of one part flush to a face of the other) then removes the remaining translational degree of freedom along that axis. Together, the two mates can reduce the pair from 6 DOF down to a single rotational degree of freedom about the shared axis (or fully fix it, if rotation is also constrained by another feature).

Q2.

Plan for roughly $1.5\text{--}2\times$ the screw diameter as the minimum thread engagement depth. A heat-set insert (a brass insert pressed in with a soldering iron) is often preferred over threading directly into printed plastic because printed plastic threads wear out and strip after repeated assembly/disassembly cycles, whereas a metal insert provides a durable, reusable thread that tolerates repeated maintenance cycles without degrading the boss.

Q3.

A compliant mechanism / living hinge is the appropriate approach — a thin, flexible section of material molded or printed as one piece with the surrounding part, allowing repeated flexing without added hardware. PP (polypropylene) and PE (polyethylene) are the two materials specifically noted as tolerating repeated flexing without fatigue-cracking.

Q4.

Reducing part count means fewer items to source, inspect, handle, and mate during assembly — each additional unique part adds handling time, potential for error, and inventory/BOM complexity. Preferring off-the-shelf components (like a standard socket head cap screw, HW-0425) avoids the cost and lead time of custom-manufacturing a part that a catalog item already satisfies, and standard hardware is typically cheaper, better documented, and easier to replace than a custom equivalent.

Q5.

An imported model with a real part number strengthens the BOM because it ties directly to a purchasable, verifiable catalog item — anyone building the assembly can order the exact part, rather than guessing at dimensions from a hand-modeled generic bolt that may not match anything actually manufactured. The limitation is that imported models are typically “dumb” solids: they carry no feature history/parametric tree, so they can be positioned and mated but not easily edited or re-parameterized within your CAD system.

Unit 4 — Design for Manufacturing (DFM)

Q1.

Flatness — Form (no datum required). Perpendicularity — Orientation (requires a datum). Position — Location (requires datum(s); position is the most commonly used location control). Profile of a surface — Profile (can be datum-related or not, depending on the callout). Circular runout — Runout (requires a datum axis).

Q2.

∅ 0.2 = a cylindrical (diametral) tolerance zone of 0.2, meaning the controlled feature's axis must fall within a 0.2-diameter cylinder. **M** = the modifier for maximum material condition (MMC), meaning the stated tolerance applies when the feature is at its worst-case (most material) size, with bonus tolerance available as the feature departs from MMC. A B C = the datum reference frame, applied in priority order: A is the primary datum (constrains the most degrees of freedom, typically contacted first), B is secondary, and C is tertiary — together the three progressively lock all 6 degrees of freedom of the part relative to the inspection/measurement setup.

Q3.

MMC (maximum material condition) is the size at which a feature contains the most material — for a hole, its smallest allowed diameter; for a shaft, its largest. LMC (least material condition) is the opposite — the least amount of material a feature can have. Bonus tolerance is granted because the total functional requirement is that a mating part must still fit/assemble; if a hole is drilled larger than its MMC (smallest) size, there is more clearance available for that mating part, so the hole's position is allowed to drift farther off-target while assembly is still guaranteed — the bonus tolerance exactly equals how far the actual size has departed from MMC.

Q4.

This is warping — caused by the part's edges and corners cooling and shrinking faster than the center of the part, causing the corners to curl. Fixes include using a brim or raft, printing in an enclosure to slow/even out cooling, and using a heated bed. This concern doesn't apply to laser cutting because laser cutting is a subtractive, sheet-based process performed on flat stock at essentially uniform temperature throughout the cut — there is no layer-by-layer thermal deposition or differential cooling of built-up material, so warping of this kind isn't a relevant failure mode.

Q5.

Cutting PVC or other chlorinated plastics on a laser cutter releases chlorine gas, which is toxic to breathe and corrosive to the machine's optics and hardware — these materials must never be laser cut. For rotating equipment (drill press, lathe, etc.), required PPE/practice includes ANSI Z87.1 safety glasses and closed-toe shoes; gloves must not be worn near a rotating spindle, chuck, or bit, because a caught glove can pull the hand into the machine.